

Module: Corporate Finance & Equity Valuation

School of Economics

University of Cape Town

Basic Valuations

Objectives

- Understand what valuations are
- Understand the terms used in the valuation of various assets
- Valuation of debentures
 - ❖ Debentures in perpetuity and redeemable debentures
- Valuation of preference shares
 - ❖ Cumulative and non-cumulative preference shares, *Our focus* **redeemable and none redeemable**, convertible and none convertible preference shares
- Valuation of ordinary shares
 - ❖ Use a zero growth model *same div. every yr*
 - ❖ Use the constant dividend growth model to value ordinary shares $\frac{D_1}{K_e - g}$
 - ❖ Apply a two stage valuation dividend model to value ordinary shares $\frac{D_0 (1+g)}{K_e - g}$ $\frac{D_1}{K_e - g}$

What are Valuations

Determination of the value of an asset stated in monetary terms

Value = present value of future cash flows

Which assets are going to be valuing?

- ❖ Debentures/Bonds
 - ❖ Preference shares
 - ❖ Ordinary shares
- } Non-redeemable + redeemable

Valuation of Debentures/ Bonds

TERMS

❖ Par Value (Face Value)

FV

❖ Coupon interest rate

C_r

❖ Maturity date (redemption date)

n

❖ Yield to Maturity (discount rate)

YTM

$$P_B = \frac{C_r}{YTM}$$

$C_r > YTM$, $P_B > FV$ - premium

$C_r = YTM$, $P_B = FV$ - par

$C_r < YTM$, $P_B < FV$ - discount

Valuation of Debentures / Bonds

- Estimate expected future cash flows – these are the “coupons” that the bond will pay plus the principal to be paid at maturity.
- Determine bond's required rate of return, or discount rate. This is the market interest rate, called bond's yield to maturity (or more commonly, yield).
- Compute current value, or price, of a bond (P_B) by calculating the present value of bond's expected cash flows:

$$P_B = C \times \frac{1 - \frac{1}{(1+r)^n}}{r} + \frac{FV}{(1+r)^n}$$

$$P_B = \frac{C}{YTM}$$

Valuation of Debentures/Bonds

Types of Bonds

Redeemable bond

- ❖ Has a fixed lifespan i.e. has a time to maturity
- ❖ Pays coupons up to the maturity date [treated as an ordinary annuity- i.e. coupon payments take place at the end of each period]
- ❖ Pays the principal at maturity [lumpsum] .i.e. face value is paid back at the of maturity.

❖ Calculation of bond value:

$$❖ P_B = C \times \underset{\text{PV of annuity}}{PVIFA}_{n,i} + F \times \underset{\text{PV of FV}}{PVIF}_{n,i}$$

$$❖ P_B = \underset{\text{m}}{C} \left[\frac{1 - \frac{1}{(1+i)^{nm}}}{\underset{\text{m}}{i}} \right] + F \left[\frac{1}{(1+i)^{nm}} \right] \quad i = \frac{i}{m}$$

Valuation of Debentures / Bonds

Redeemable debentures / bonds

Face value is repaid on the set maturity date, plus periodic coupon amount until that date.

General Formula - **using Tables and a calculator**

$$\left(\begin{array}{c} \text{Price} \\ \text{of bond} \end{array} \right) = \left(\begin{array}{c} \text{present value} \\ \text{of coupon payments} \end{array} \right) + \left(\begin{array}{c} \text{present value} \\ \text{of face value at maturity} \end{array} \right)$$

Specific valuation formula, using tables and a calculator.

$$\text{Price bond} = (C \times \text{PVIFA } i\%, n \text{ periods}) + (F \times \text{PVIFA } i\%, n \text{ periods})$$

where:

- C is the coupon per period = $\overset{FV \times Cr}{\text{face value} \times \text{coupon interest rate}}$
- F is the face value ^{FV} on maturity (or face value)
- i is the discount rate which is given or stated as the ^{YTM} yield to maturity

Valuation of Debentures / Bonds

Example 1

A three year bond of R1000 pays an annual coupon of 8%. The market rate of interest is 10%

You can use a time line, Calculator or Formula to solve for the price of the bond.

Time line: coupon amount per year = $0.08 \times R1000 = R80$

	0	1	2	3
		R80	R80	R80
		$\frac{R80}{(1.1)^1}$	$\frac{R80}{(1.1)^2}$	$\frac{R80}{(1.1)^3}$
R72.73	←			
R66.11	←			R1 000
R60.11	←			$\frac{R1\ 000}{(1.1)^3}$
R751.30	←			
$P_B = R950.25$				

∴ Trading @ discount

Valuation of Debentures / Bonds

Using Financial Tables

❖ The use of a timeline in the previous slide involved 4 calculations but this could be reduced to two calculations if we use a formula.

$$P_B = (C \times PVIFA \text{ } i\%, n \text{ periods}) + (F \times PVIF \text{ } i\%, n \text{ periods})$$

R1000 x coupon 8%



$i = 10\%$ $n = 3$



$$P_B = (R80 \times PVIFA \text{ } 10\%, 3 \text{ years}) + (R1000 \times PVIFA \text{ } 10\%, 3 \text{ years})$$

$$P_B = (R80 \times 2.4869) + (R1000 \times 0.7513)$$

$$P_B = (R198.95) + (R751.30) = R950.25$$

Valuation of Debentures / Bonds

Using Formula *m = 1 ∴ no adjustments*

$$\diamond P_B = C \left[\frac{1 - \frac{1}{(1+i)^{nm}}}{i} \right] + F \left[\frac{1}{(1+i)^{nm}} \right]$$

$$\diamond P_B = 80 \left[\frac{1 - \frac{1}{(1+0,10)^3}}{0,1} \right] + 1000 \left[\frac{1}{(1+0,1)^3} \right]$$

$$\diamond P_B = 198,94 + 751,31$$

$$\diamond P_B = R950,25$$

Valuation of Debentures/Bonds

- Example 2

Assume that Padua Limited issued a bond with a face value of R1000^{FV} and a coupon rate of 15%^{C_r}. The bond pays coupons on a semi-annual^{m=2} basis and it is redeemable in 5 yearsⁿ⁼⁵ time. Similar bonds have a market yield of 10%^{YTM}.
expect $P_B > FV$ because $C_r > YTM$

- Required

Calculate the price that will be paid on this bond today

$$P_B = \frac{150}{2} \times \frac{1 - \frac{1}{\frac{1,15}{2}^{5 \times 2}}}{\frac{0,1}{2}} + \frac{1000}{1,15^{5 \times 2}}$$

$$= R 1193,04$$

∴ Trading @ premium

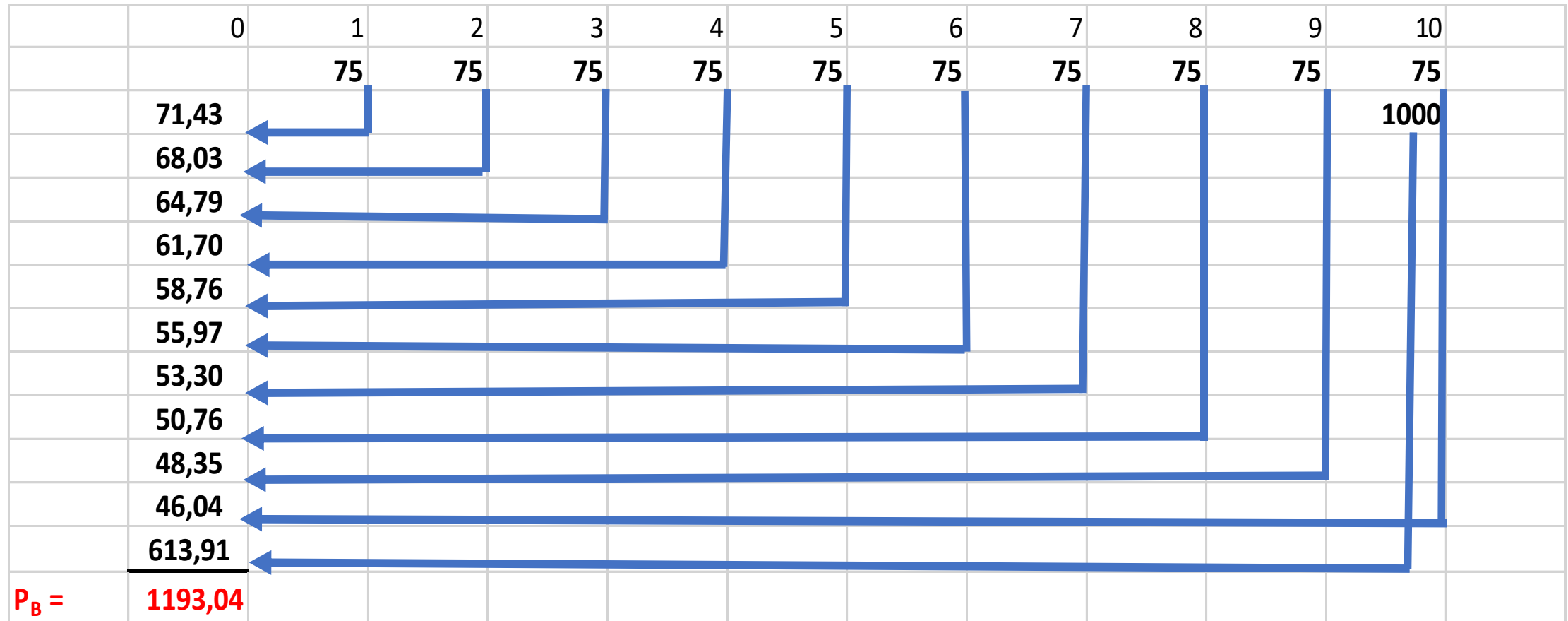
Valuation of Debentures/Bonds

Example 2 Solution

- **Hint:** Before you start doing your calculation, note that your coupon payments are made more than once per year. Given that, therefore, you need to adjust three things;
- **First, Adjust your coupon rate.** that is your annual coupon rate should be adjusted to reflect a periodic rate, in our question above, you divide the annual coupon rate by 2 since its a semi annual coupon bond (meaning coupons are paid twice per year). That means, you divide your annual coupon rate by 2 first and then multiply the result by the face value to get your periodic coupon amount. OR you can multiply the given annual coupon rate by the face value and get the annual coupon amount, but remember to divide the annual coupon amount you get by the number of payment periods per year as suggested by the question, in our case divide the annual coupon amount by 2 to get the semi annual coupon amount.
- **Second, adjust your YTM** also to match the coupon payments, so in the case of the question we have at hand, you divide your YTM by 2 as well.
- Third, **adjustment number of years** under consideration to reflect the total number of payments over the entire number of years in the question, in this case its 5 but 2 payments per year, hence, total number of payments should be 5 multiply by 2 for this question (i.e. nm ; m is number of payments per year, and n is the number of years in the question). In the case above $nm = 5 \times 2 = 10$.

Valuation of Debentures/Bonds

- Example 2 Solution- using the time line



Valuation of Debentures/Bonds

Example 2 solution- Using Formula

$$\diamond P_B = C/2 \left[\frac{1 - \frac{1}{(1+i/2)^{nm}}}{i/2} \right] + F \left[\frac{1}{(1+i/2)^{nm}} \right]$$

$$\diamond P_B = \frac{R150}{2} \left[\frac{1 - \frac{1}{(1+0,10/2)^{10}}}{\frac{0,1}{2}} \right] + R1000 \left[\frac{1}{(1+0,1/2)^{5 \times 2}} \right]$$

$$\diamond P_B = R75 \left[\frac{1 - \frac{1}{(1+0,05)^{10}}}{0,05} \right] + R1000 \left[\frac{1}{(1+0,05)^{10}} \right]$$

$$\diamond P_B = R579,13 + R613,91$$

$$\diamond P_B = \mathbf{R1193.04}$$

Valuation of Debentures / Bonds

Redeemable debentures / bonds

Example 2 Solution using financial tables

$$P_B = (C \times PVIFA \text{ } i\%, n \text{ periods}) + (F \times PVIF \text{ } i\%, n \text{ periods})$$

$$\text{PV of interest: } R75 \times 7,7217 = R579,13$$

$$\text{PV of redemption of face value: } R1000 \times 0.6139 = R613,90$$

$$\text{Total value} = R579,13 + R613.91 = R1193,04$$

$$\text{Therefore, } P_B = R1193,03$$

[illegible]

Valuation of Debentures/Bonds

- Example 3

Assume that Padua Ltd issued a 30 year bond 15 years ago. The bond has a par value of R1000 and it has a 10% coupon rate which is paid semi annually.

Handwritten notes: $n = 30 - 15$, FV , $Cr = 5\%$, $m = 2$

Required

- Calculate the price of the bond today assuming similar bonds are trading at a yield of 10%.
 - Calculate the Value the bond assuming a yield to maturity of 8%.
 - Calculate the Value the bond assuming a yield to maturity of 12%.
- Handwritten notes for (a):* P_0 , $YTM = 5\%$, expect par because $Cr = YTM$
- Handwritten notes for (b):* prem
- Handwritten notes for (c):* discount

Valuation of Debentures/Bonds

- Example 3 solution

Hint: Adjustments need to be done to the inputs into the calculation

Face Value = R1000

Coupon rate 10% paid semi annually, thus periodic coupon rate = $10\%/2 = 5\%$

TYM for (a) is 10%, there adjustment to match coupon payments = $10\%/2 = 5\%$

TYM for (b) is 8%, hence, adjustment = $8\%/2 = 4\%$

YTM for (c) is 12%, hence, adjustment = $12\%/2 = 6\%$

On issue, the bond had 30 years, but now 15 years have gone by, meaning that the number of years remaining for the bond to mature = 15, thus $n = 15$, hence, adjustment is $nm = 15 \times 2 = 30$ total payment periods

Note that: Coupon rate is fixed at the time of issue and remains fixed for the entire time to maturity of the bond while the yield to maturity varies according to changes in the market interest rates. Practically the coupon rate and the YTM are usually set at the same rate (for bond issued at par, i.e. price of the bond and face value being equal), but as soon as the bond starts trading, the YTM starts to vary and differ from the coupon rate.

However, there are situations where the bond can be issued at a discount (price below face value) or issued at a premium (i.e. with price above the face value).

In the example above coupon rate is fixed at 10% paid semi annually, therefore, the periodic coupon amount = $0,05 \times 1000 = R50$

Valuation of Debentures/Bonds

Showing the relationship between yield to maturity and bond price

Example 3 solution continued.

- a) Calculate the price of the bond today assuming similar bonds are trading at a yield of 10%.

$$P_B = R50x \left(\frac{1 - \frac{1}{(1 + 0.05)^{30}}}{0.05} \right) + [R1,000x \left(\frac{1}{(1 + 0.05)^{30}} \right)] = R1,000$$

- b) Value the bond again assuming a yield of 8%

$$P_B = R50x \left(\frac{1 - \frac{1}{(1 + 0.04)^{30}}}{0.04} \right) + [R1,000x \left(\frac{1}{(1 + 0.04)^{30}} \right)] = R1,172.92$$

- c) Value the bond again assuming a yield of 12%

$$P_B = R50x \left(\frac{1 - \frac{1}{(1 + 0.06)^{30}}}{0.06} \right) + [R1,000x \left(\frac{1}{(1 + 0.06)^{30}} \right)] = R862.32$$

Note inverse relationship btwn P_B & YTM

$$P_B = \frac{C}{YTM}$$

When similar bonds start trading to yield ^{YTM} 20%, investors now expect this level of return, hence they will demand 20% return on every similar bond on the market.

- If the bond continues to trade at a price of R100 no one will buy it. Or if the bond continues to trade at R123.33 no one will buy it because that price will give a yield of 9%.
- Investors will now start discounting the cash flows at the new RRR hence bringing down the price
- And how does the price adjust. You simply discount the cash flows of that bond using the required rate of return.
- So the price decreases because you are now discounting the cash flows at a higher rate

NB: After doing the calculations, for each question compare the 1. ^{FV}Face Value to the ^{P_B}Price and
2. the ^{C_r}Coupon rate to the ^{YTM}YTM

Valuation of Debentures/Bonds

- Example 3 - Observations from the calculations in the previous slide.
- There is an inverse relationship between the Bond's price and yield to maturity
- If the ^{C_r} Coupon rate = ^{YTM} yield to Maturity, the bond's price and face value are equal, hence, bond is trading at par [e.g (a) where the coupon rate of 10% = YTM of 10%]. ^{$P_b = FV$}
- If the Coupon rate > yield to Maturity, the bond's price is greater than the face value, hence, bond is trading at premium [e.g. (b) where the coupon rate of 10% > YTM of 8%]
- If the Coupon rate < yield to Maturity, the bond's price is less than the face value, hence, bond is trading at a discount. [e.g.(c) where the coupon rate of 10% < YTM of 12%]

Valuation of Debentures / Bonds

Debentures in perpetuity (non-redeemable debentures)

This is a bond that has no redemption date

Example 4

Assume that a 15% coupon bond with a Face value of R1000 was issued in perpetuity
Further assume that the bond's required return is 9%. *∴ premium*

Required

Calculate the value of the bond

$$P_B = \frac{C}{YTM}$$

Example 4 Solution:

$$P_B = C/I$$

$$C = 0,15 * R1000 = R150$$

$$I = 0,09$$

$$P_B = R150/0.09$$

$$P_B = R1666.67$$

Valuation of Debentures/Bonds

Risks associated with bonds

Interest rate risk

- ❖ Relates to the uncertainty about future bond values that is caused by the volatility or fluctuations of interest rates/YTM.

Default risk (corporate bonds)

- ❖ The probability that the bond issuer or borrower can fail to meet any condition of the debt or loan contract constitutes default

Valuation of Preference shares

Preference shareholders receive a fixed dividend whilst debenture (bond) holders receive interest.

Dividends are paid only after all expenses have been incurred, including interest. This means that dividends are more risky and therefore preference shares **need to offer a higher return.**

- What is the effect of taxation? -> ^{div paid after tax} does not go through the income statement, hence, does not affect taxable income

Types of preference shares

- ❖ Cumulative and none cumulative
- ❖ Redeemable ^{to maturity} and none redeemable ^{indefinitely}
- ❖ Participating
- ❖ Convertible and none convertible

Valuation of Preference shares

To determine the intrinsic value of preference shares,

- Discount all the future cash flows (dividends) using the required rate of return (dividend yield for similar preference shares) as the discount rate. $RRR = i$

$$P_p = PV(\text{dividend payments})$$

Valuation of preference shares

Preference share in **perpetuity/Non-redeemable** preference shares

Pay an infinite series of equal and periodic cash flows. Valued using the perpetuity formulae as follows.

$$P_p = \frac{D_p}{i = RRR}$$

- Example 5

Assume that many years ago, Padua Ltd. issued preference shares that pay an annual dividend equal to R6.80^{*D_p*} every year in perpetuity. If the required rate of return on similar risk investments is 8 percent^{*i*}. What should be the market value of the Padua Ltd's preference shares?

Example 5 Solution

$$P_p = \frac{D_p}{i} = \frac{R6.80}{0.08} = R85$$

Valuation of Preference shares

Redeemable preferences shares

Example 6

- Determine the value of 10,000 redeemable preference shares. The preference shares have a par value of R1 each and they pay a dividend of 8%. The required rate of return on similar preference shares (dividend yield) is 10%. The shares are redeemable in full at par value after 5 years.

Example 6 Solution (priced similar to a bond)

$$\begin{aligned} P_p &= Dx \left(\frac{1 - \frac{1}{(1+i)^n}}{i} \right) + Px \left(\frac{1}{(1+i)^n} \right) \\ &= 10000 \times \frac{1 - \frac{1}{(1,1)^5}}{10\%} + \frac{10000}{1,1^5} = R9\,241,84 \end{aligned}$$

Valuation preference shares

- Example 6 Solution

$$P_p = D \times PVIFA_{5,10\%} + P \times PVIF_{5,10\%}$$

$$P_p = 800 \times 3.791 + 10,000 \times 0.621$$

$$P_p = R9,242.80$$

Valuation of ordinary shares (equity)

The valuation of ordinary shares is viewed as more challenging compared to valuation of fixed income instruments. The reasons behind that are:

- ❖ the promised cash flows (.i.e. dividends) are not known in advance
- ❖ there is no way to easily observe the required rate of return by the equity investors.

Valuing Ordinary Shares - need expected div

- The Dividend Discount Model (DDM)

- ❖ Based on the DDM, the intrinsic value of equity should be equal the present value of all expected future dividends to perpetuity

- ❖ General Formula

$$P_0 = PV(\text{All future dividends})$$

Steps in the Valuation process using the Dividend Discount Model

1. Determine/Project future dividends of the firm
2. Determine the Required Rate of Return (RRR)- which is the cost of equity K_e
3. Discount the dividends **to the valuation date** at the required rate of return (i.e. cost of equity) $RRR = K_e$

Valuation of ordinary shares

DDM

- Versions of the Dividend Discount Model

Zero growth in dividend $\frac{D}{K_e - g}$

- ❖ Assume a company is expected to maintain the same dividend in future years

Constant growth model/Gordon/Dividend Growth Model

- ❖ Assumes that dividends are growing at a constant rate

Non constant growth model/2 stage model

- ❖ Assumes high growth rates over a given period^①, after which dividends are expected to grow at a sustainable growth rate^② into the foreseeable future.

$$\frac{D(1+g)}{K_e - g} + \frac{D}{K_e - g}$$

① ②

Valuation of ordinary shares

- Zero growth model

Assume a company is expected to maintain the same dividend in future years. What are we valuing?

It is similar to valuing a perpetuity.

$P_0 = D / k_e$, where D is the dividend amount and k_e is cost of equity

Example 7 and solution

So, If dividend is R0.80 and the Cost of Equity is 11%, then the value of the ordinary share is $0.80 / .11 = R7.27$

$$P_0 = \frac{0.8}{0.11} = R7.27$$

Valuation of ordinary shares

Dividend Growth Model (constant growth model)

- ❖ Value of an ordinary share = PV of future dividends
- ❖ If dividends are growing at a constant rate, then;
- ❖ **$PV = D(1+g)/(k_e - g)$ or $PV = D_1 / (K_e - g)$**
 - Where
 - D_0 ^{this yr} $(1+g)$ = next year's dividend i.e. D_1
 - k_e = cost of equity
 - g = future constant growth rate in dividends

What are valuing? - -> a growing perpetuity

Valuation of Ordinary shares

Dividend growth model

- How do we estimate growth in dividends?

❖ Use surrogates/proxies such as;

- Growth in earnings
- Growth in cash flow
- Sustainable growth rate formula as a check .i.e $[\text{ROE} \times (1 - \text{payout ratio})]$ *retention ratio*

Growth in dividends over the long term should reflect earnings growth. If EPS is growing at 10% per year, and if current dividend growth is 20%, then this is **not sustainable over the long term**. Earnings growth is a long-term anchor for dividends.

Valuation of ordinary shares

Constant growth Dividend Discount Model ^{DDM}

Example 8 (given expected dividend)

Assume that Harvey Ltd expects to pay R2.15 dividend in the next period. Assume that dividends have been increasing at 6% per annum and are expected to continue at this rate in the future. ^{D_1} ^{g}

Required

Calculate the maximum price you will be willing to pay for this stock if your required rate of return is 11%.

- Example 8 solution ^{$RRR = K_e$}
- $P_0 = \frac{D_1}{k_e - g}$ ^{growing perpetuity formula}
- $P_0 = \frac{2.15}{0.11 - 0.06}$
- $P_0 = R43$

Valuation of ordinary shares

- Constant growth Dividend Discount Model (given a past dividend D_0)

Example 9

Assume that Porsh'e Ltd paid R3 dividend this year and that the firm expects the dividends to grow at a constant rate of 5% each year forever. Further assume that Investors require a 12% return to invest in this stock.

Required

- What is the intrinsic price of this share.

$$P_0 = \frac{D_0 (1+g)}{K_e - g} = \frac{3 (1,05)}{0,12 - 0,05} = R 45$$

Valuation of ordinary shares

D_0

- Constant growth Dividend Discount Model (given a past dividend)

Example 9 solution

Things to note:

- Here you are given a historical dividend but our calculation should be based on future dividends, that is expected dividends. Therefore we need to calculate D_1 , which is given as: $D_1 = D_0 \times (1+g)$

- In this case, $D_1 = 3 \times (1,05) = \text{R}3,15$

- $P_0 = \frac{D_1}{ke-g}$ or $P_0 = \frac{D_0 \times (1+g)}{ke-g}$

- $P_0 = \frac{3,15}{0,12-0,05} = \text{R}45,00$ or $P_0 = \frac{3 \times (1,05)}{0,12-0,05} = \text{R}45,00$

Valuation of ordinary shares

Constant Dividend Discount Model (given historical dividends).

Example 10

- Assume that ABC Ltd paid the following dividends over the last 5 years $R1.00, R1.03, R1.09, R1.10, R1.17$. In addition, assume that future dividends are expected to grow at the same historical compound rate. The required rate of return on this shares is 11%.
 PV $FV = D_0$ k_e

Required

- Calculate the value of the ABC ordinary share.

Valuation of ordinary shares

Example 10 solution

- First of all you need to calculate “g” and this is done as below:

n = no. of observations - 1 = no. of periods

- $g = \left[\left[\frac{FV}{PV} \right]^{\frac{1}{n}} - 1 \right]$

- Given the information in the previous slide, we have:

- $g = \left[\left[\frac{R1.17}{R1.00} \right]^{1/4} \right] - 1 = 4\%$

- Thereafter we can now calculate value of a share, using last dividend paid as D_0

- $P_0 = \frac{1,17 \times (1,04)}{0,11 - 0,04} = R17,38$

Valuation of ordinary shares

DDM

Dividend Discount Model (2 Stage/Non-Constant growth model)

This is mostly suitable for companies that are experiencing a high growth in dividends which is followed later by a lower but stable growth rate in dividends.

$$\diamond P_0 = PV (\text{future dividends})$$

$$\diamond P_0 = PV (\text{Mixed/abnormal dividend growth}) + PV (\text{Constant dividend growth})$$

Valuation of ordinary shares

2 Stage/None constant Dividend Discount Model

Example: 11

Assume that Gitsalina Ltd is **expected** to experience a growth rate of 30%^g for the next 3 yearsⁿ⁼³ and thereafter will experience a growth rate of 6%^g per year. The cost of equity is 12%^{Ke} and the current dividend is 60^{D₀} cents.

Required

Calculate the Value of the Gitsalina share

$$P_0 = \frac{D_0 (1+g)}{K_e - g}$$
$$= \frac{0,6 (1,3)}{0,12 - 0,06}$$

Valuation of ordinary shares

- Example 11 Solution, 2 stage/None constant Dividend Discount Model

	0	1	2	3
Expected dividends		^{D₁} 0,78	^{D₂} 1,014	^{D₃} 1,3182
Continuing Value $(D_3 \times (1+g)/(k_e-g))$				23,2882
Total cash flows		0,78	1,014	24,6064
Discount Factors @12% ^{$(1+12\%)^n$}		0,8929	0,7972	0,7118
PVs of Cash Flows		0,6964	0,8084	17,5143
Sum of PVs of cash flows .i.e ^{Intrinsic} Value of share	19,02			

Valuation of ordinary shares

- 2 Stage/None constant Dividend Discount Model
- Now assume a situation whereby a company is growing at a rapid rate in terms of market share and will only pay a dividend in the distant future.
- Example 12
- Assume that a biotech company expects to only start making a profit from year 8 and expect to pay a dividend of R0,20^{D₁₁} in year 11, which will double until year 13 and then grow at 7.5%^g per year.
- Required
- (a) Calculate the value of a share assuming that the cost of equity is 14%^{K_e}
- (b) Calculate the value of a share assuming that the cost of equity decrease by 3%, holding everything else the same.

^{K_e 11.1%}

Example 12 Solution part (b)

Valuation of ordinary shares

- 2 Stage/None constant Dividend Discount Model

Example 13

Let's apply the dividend growth model to Gigitsali Ltd

Assume a long-term sustainable growth rate of 8% (approximately 4.5% inflation and 3.5% real growth). Assume the cost of equity is 12.2% (Risk free rate of 7.2% + risk premium of 5%).

Assume that the high growth in earnings and dividends will continue for the next 5 years, before falling to 8% per year. Expected growth rate for 5 years is 15% per year, which reflects the growth in earnings and dividends 2013-2019. Assume that the current dividend is R0,944.

Required

Calculate the value of a Gigitsali share

Valuation of ordinary shares

- 2 Stage/None constant Dividend Discount Model

Example 13 Solution

	0	1	2	3	4	5
Expected dividends		1,086	1,248	1,436	1,651	1,899
Continuing Value $(D_5 \times (1+g))/(k_e-g)$						48,824
Total cash flows		1,086	1,248	1,436	1,651	50,723
Discount Factors @12,2%		0,8913	0,7944	0,7080	0,6310	0,5624
PVs of Cash Flows		0,967558	0,991704	1,016452	1,041818	28,52597
Sum of PVs of cash flows .i.e Value of share	32,54					

Valuation of ordinary equity

2 stage/None constant Dividend Discount Model

Example 14

Assume that an investor is analyzing Kulisana Ltd with the objective to invest in the company. Further assume that Kulisana Ltd has just paid a dividend of R7. D_0

In addition, assume that the investor's projections show that Kulisana Ltd's dividends for the next 3 years are expected to grow at a rate of 30%,^g after which all future dividends after the 3rd are expected to grow at a rate consistent with a return on equity of 10% and a dividend payout ratio of 60%.

Required

$$\begin{aligned} g &= ROE \times (1 - \text{payout ratio}) \\ &= 10\% \times (1 - 60\%) = 4\% \end{aligned}$$

Calculate the value of the Kulisana Ltd share if the cost of equity is 14% ^{K_e}

Valuation of ordinary shares

- 2 Stage/None constant Dividend Discount Model

Example 14 Solution

First of all calculate g

$$g = \text{ROE} \times (1 - \text{payout ratio}) = 10\% \times (1 - 0,6) = 4\%$$

	0	1	2	3
Expected dividends		9,10	11,83	15,38
Continuing Value $(D_3 \times (1 + g) / (k_e - g))$				159,94
Total cash flows		9,10	11,83	175,32
Discount Factors @14%		0,8772	0,7695	0,6750
PVs of Cash Flows		7,98	9,10	118,34
Sum of PVs of cash flows .i.e Value of share	135,42			

Valuation of ordinary shares

- Some of the tricks that you need to know whether for the exam or assignment are that you might be given any other information that might help you to calculate dividends per share. The tricks that are going to be relevant for you are as shown below. Familiarise yourselves with them, so that you get to know how to get your dividends per share given such information.

$$g = ROE \times (1 - \text{pay-out ratio})$$

You might be given the total earnings, dividend pay-out ratio and number of issued shares.

So you may need to remember the following:

✓ **Total Dividends = Earnings x pay-out ratio**

$$\text{Payout ratio} = \frac{\text{DPS}}{\text{EPS}}$$

✓ **Dividend per share ^{DPS} = total dividends/number of shares**

You can be given the EPS and the dividend pay-ratio

✓ **To solve for the DPS = EPS x Payout ratio**

Or you can be given the EPS and the dividend cover ratio

✓ **To solve for the DPS = EPS/Dividend cover**

Valuation of ordinary shares. Example 15

BioTech Ltd is a young company in the bio-technology industry and expects to have a strong growth rate in earnings and dividends over the forthcoming four⁴ years, after which the rate of growth will settle at a long term sustainable rate estimated at 8%.^g You believe these estimates are realistic. For the financial year ended 31 December 2019, the company posted earnings per share of 280^{D₀}cents. In the year 2019, BioTech paid out 40%^{payout ratio} of its earnings as dividends. It is anticipated that this pay-out ratio will be maintained in the future. Share analysts have forecast the following earnings per share growth over the next four years:

Year	2020	2021	2022	2023
Earnings growth	25%	20%	16%	12%

The company has net assets^{f_m} (i.e. equity) of R89.6m and has 10million shares in issue. BioTech has a beta of 1.5, the market return is 12%, and the return on a government security is 6%.^{r_f} BioTech's shares are currently trading at PE ratio of 10.5 times. Five years ago BioTech issued non-redeemable preference shares. The preference shares pay a dividend amount of R160,000 and the dividends are payable annually on the 31st of December. The market yield for preference shares of a similar type and risk profile is 11%.^B

Required

1. Calculate the cost of equity^{K_e} of BioTech Limited from the information given in the question above.
2. Calculate the value of one ordinary share of BioTech Limited using the Dividend Discount Model^{DDM} as at 31 December 2019.

Valuation of ordinary shares

- 2 Stage/None constant Dividend Discount Model

Example 15 Solution. Note that 2019 is the valuation date.

First of all calculate cost of equity

$$K_e = R_f + \text{Beta} (R_m - R_f) = 6\% + 1,5 * (12\% - 6\%) = 15\%$$

	2019	2020	2021	2022	2023
Current earnings per share (.i.e. EPS_0)	2,8				
growth in earnings per share		25%	20%	16%	12%
Expected earnings per share		3,50	4,20	4,87	5,46
Payout ratio		40%	40%	40%	40%
Expected dividends per share		1,40	1,68	1,95	2,18
Continuing Value $(D_{23} \times (1 + 0,08)) / (0,15 - 0,08)$					33,68
Total cash flows		1,40	1,68	1,95	35,86
Discount Factors @15%		0,8696	0,7561	0,6575	0,5718
PVs of Cash Flows		1,22	1,27	1,28	20,50
Sum of PVs of cash flows .i.e Value of share	24,27				